

Your first Python research experiment.

Can data help us understand
what happens on the soccer field?

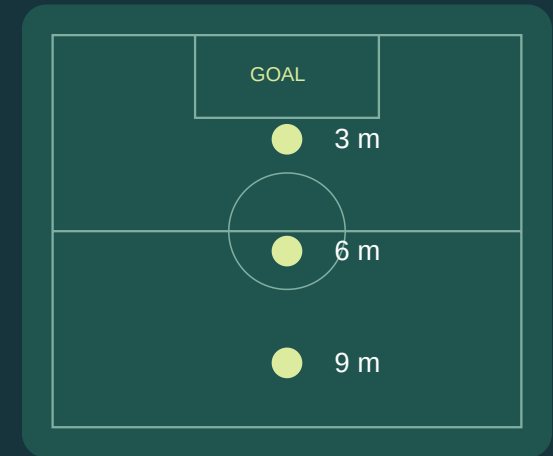
PREDICT

CODE

EXPLAIN

Bo Shang · 50-minute trial lesson

No account or installation needed. A keyboard helps with the code activities.



×



恒点学术
MeritPoint
Academy

Students first. Growth for the long term.



MeritPoint Academy 恒点学术

Research education for secondary-school students,
connecting AI, interdisciplinary projects and guided practice.

全心全意为学生服务

Start with the student

学生第一 · 因材施教

Match the next step to each student's
interests and readiness.

Learn through real work

真实探究 · 学术诚信

Students understand and explain their
own questions, code and evidence.

Build lasting ability

长期成长 · 持续支持

Develop independent thinking through
practice, guidance and feedback.

[Our mission & learning values · 办学宗旨与培养理念](#) ↗

Built around what students need.

AROUND 2025 约 2025 年

AS WE GREW 随后逐步成型

SUMMER 2026 2026 年暑期

FALL 2026 2026 年秋季

01

Community beginnings

社区公益科研辅导

Xiwang Guo (NJIT) began free Sunday research mentoring at Livingston Library.

郭希望老师，周日下午 1-4 点，陪伴高中生探索科研。

02

One-to-one research

系统化一对一科研指导

Guidance from a research question to data, evidence and writing.

部分学生取得论文录用；逐步形成约 15 次 × 1 小时的系统指导。

03

AI Toolbox small groups

AI 项目学习走进小班

Bo Shang led two cohorts: around 15 lessons, up to 5 students per group.

尚博老师带领，连接 AI 工具、项目实践与科研方法。

04

Python group classes

正式开展 Python 团课

Following limited one-to-one summer pilots, a group entry point for beginners.

把支持延伸到编程起点，让学习路径更加完整。

One mission. A fuller learning path.

从学生需要出发，逐步补齐每一段学习支持。

Python → AI Toolbox → Research

Program milestones · based on the team's account · [团队口述沿革 ↗](#)

AI expands the toolkit. Students lead the research.

STUDENT · 学生

**Ask. Decide.
Explain.**

提出问题，作出判断，解释证据。

AI · 工具

**Explore. Code.
Analyze.**

辅助构思、编程与分析；使用须披露、结果须核查。

MENTOR · 导师

**Guide. Question.
Review.**

提供方法指导、反馈与关键节点审查。

Question

提出问题



Data & code

数据与代码



Evidence

检验证据



Explain

表达与反思

Today: run it, change it, and explain why.

今天就用一个小实验，亲自走一遍这个过程。

[Our research & AI-use standards · 科研与 AI 使用原则](#) ↗

Meet the **mentor team**.



TODAY'S INSTRUCTOR
LAB DIRECTOR · 项目负责人

Bo Shang, PhD

Postdoctoral Associate
University of Maryland

马里兰大学博士后研究人员

Robotics · Multi-sensor perception
Computer vision · Machine learning

机器人 · 多传感器感知 · 计算机视觉 · 机器学习



Professor Zhu 朱教授

Software engineering · Collaborative & multi-agent systems

软件工程 · 协同与多智能体系统



Professor Guo 郭教授

AI · Machine learning · Intelligent optimization

人工智能 · 机器学习 · 智能优化



Dr. Wang 王博士

AI strategy · Management education · Foresight

AI 战略 · 管理教育 · 技术前瞻



Dr. Li 李博士

AI model governance · Finance & risk

AI 模型治理 · 量化金融与风险

Mentors and advisors are matched by project and availability.

导师与顾问按课题方向和时间安排匹配；以上介绍沿用官网公开范围。

[Meet the team · 查看导师团队 ↗](#)

From Python to **your own research.**

STAGE I · 第一阶段

01

Python for AI Research

AI 科研的基础能力课程

- Python & data analysis
- NumPy · pandas · Matplotlib
- Scientific computing & AI coding basics

A foundation for AI research

从零基础开始，建立编程与数据能力

STAGE II · 第二阶段

02

AI Toolbox

AI 科研能力训练营

- Machine learning · Deep learning
- LLMs · Computer vision
- Research methodology & AI projects

Build projects. Learn the research process. Read papers.

完成 AI 项目，掌握流程，学习阅读论文

STAGE III · 第三阶段

03

AI Research Mentorship

AI 科研导师计划

- Doctoral mentors · Individual research
- Question · Literature · Model design
- Experiments · Analysis · Paper writing

Portfolio · Poster · Presentation

在导师指导下，完成并表达自己的科研项目

YOU ARE HERE · 今天

Stage I: Python for AI Research

先用一个小实验，练习编程、数据分析与证据表达。

Pathway for Grades 8-12 · Beginners can start in Stage I. [MPA program outline](#) ↗

Python: from code to data.

LESSONS 1-3

UNIT 1 / 4

Build the language

学会表达规则

- 01 Your First Python Experiment
- 02 Decisions & Loops
- 03 Functions & Research Records

YOU BUILD · 学习成果

A program you can explain + your question card

能解释的小程序，以及自己的研究选题卡

LESSONS 4-8

UNIT 2 / 4

Build the evidence

用数据检验想法

- 04 NumPy & Simulation
- 05 pandas & Tables
- 06 Cleaning With a Reason
- 07 Charts That Answer Questions
- 08 Exploring a Research Question

YOU BUILD · 学习成果

Documented data + a chart + a test plan

有来源的数据、能说明问题的图表和实验计划

TODAY · 今天

Try Lesson 1: predict, run, change and explain.

今天先体验第 1 课：预测、运行、修改，再解释结果。

Python: from models to research.

LESSONS 9-13

UNIT 3 / 4

Compare fairly

建立基线，公平比较

- 09 Your First Machine-Learning Baseline
- 10 Predicting Numbers
- 11 Predicting Categories
- 12 Trees & Overfitting
- 13 A Fair Model Comparison

YOU BUILD · 学习成果

A baseline + actual results + explained errors

基线与真实对照结果，解释失败和局限

LESSONS 14-15

UNIT 4 / 4

Explain and share

整理证据，表达研究

- 14 A Reproducible Research Story
- 15 Project Showcase

YOU BUILD · 学习成果

Runnable notebook · Figures · Abstract & poster drafts

可重跑项目、图表、摘要与海报草稿，以及展示

ONE PROJECT

One question grows into a project you can run, check and explain.

围绕一个小问题逐步积累证据；Python 中的入门建模为后续 AI Toolbox 打基础。

Make something you can explain.

01 · A PROGRAM

A few lines of Python

Run them. Make one meaningful change.

02 · A CHART

One clear comparison

Turn numbers into something you can see.

03 · A RESEARCH NOTE

One finding + one limit

Use a result from your own experiment.

YOUR TURN

Work with a partner: one predicts, one runs. Then switch.

What can Python do?



Explore data

数据分析

Compare records. Turn numbers into a chart.



Build AI models

AI 模型

Predict a temperature, then check the errors.



Work with images

图像处理

Locate a ball in an image.



Games & simulations

游戏与模拟

Try a movement rule in a maze.



Automate tasks

自动处理重复任务

Organize a folder of files with a repeatable rule.



Power web tools

网页工具

Calculate a result behind a website.

Which would you like to explore? Pick a card.

你最想探索哪一个？点选一张图卡。

TODAY · 今天

One small data experiment: **soccer shots** → **Python** → **evidence**.

应用有很多；今天先用足球数据，完成一个能解释的小实验。

[Application examples & sources ↗](#)

Does scoring change with distance?

Which distance do you think will have the highest goal rate?

A

Closer shots

3 metres will have the highest rate.

B

About the same

The three rates will be similar.

C

I'm not sure yet

I want to inspect the shots first.

Choose an idea. You can change your mind when you see the evidence.

A prediction is a starting idea, not a grade. Your choice stays in this browser.

All shot records in this lesson are invented teaching data, not a real player study.

Your browser is a tiny research lab.

CODE → RUN → OUTPUT

Everything happens here.

Read the short program.
Predict what it will display.
Press **Run Python** and check.

Prepare Python

The first load downloads the Python runtime.

YOUR CONTROLS

Keep the tools close.

Listen explains the current slide.
Explain code walks through a starter.
Reset code gives you a fresh start.

Use the arrow buttons to move through the lesson. On a small screen, scroll the slide.

YOUR TURN

Ready when you are. Preparing Python should not require a sign-in.

Make Python say something.

Change the words inside the quotes. Keep the quotes and parentheses.

experiment.py

Python · editable

```
print("Hello, research!")
```

YOUR OUTPUT

Predict first. Then run your code.

YOUR TURN

Run it once. Change the message. Run it again.

Give the numbers a name.

A variable is a name that holds a value. Here, attempts counts every shot.

experiment.py

Python · editable

```
goals = 3
attempts = 5
rate = goals / attempts * 100
print(rate)
```

YOUR OUTPUT

Predict first. Then run your code.

YOUR TURN

Before running: what is $3 \div 5 \times 100$?

A list remembers every attempt.

1 = **goal** • 0 = **miss**. Each position is one attempt in the 6 m group.

experiment.py

Python · editable

```
shots = [1, 0, 1, 1, 0]
print(sum(shots))
print(len(shots))
print(sum(shots) / len(shots) * 100)
```

YOUR OUTPUT

Predict first. Then run your code.

YOUR TURN

Point to the code that counts goals. Then point to all attempts.

Compare the same kind of number.

3 m

5 shots

**4 / 5** goals · 80%

6 m

5 shots

**3 / 5** goals · 60%

9 m

5 shots

**2 / 5** goals · 40%

●1 = goal ●0 = miss

Synthetic data made for this lesson.

3 m: 4 of 5 = 80%. 6 m: 3 of 5 = 60%. 9 m: 2 of 5 = 40%.

Turn the table into a picture.

The chart is generated by Python. The first chart run also loads the plotting library.

experiment.py

Python · editable

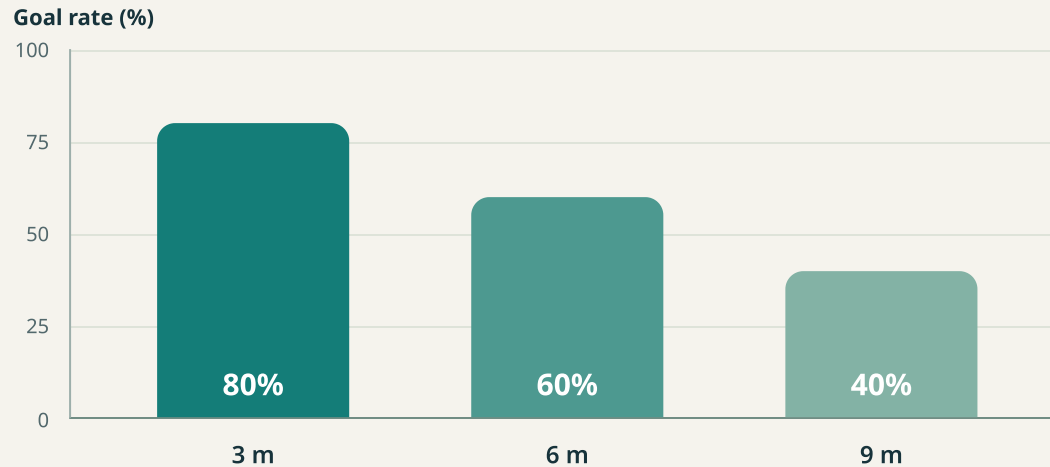
```
import matplotlib.pyplot as plt
near = [1, 1, 1, 1, 0]
mid = [1, 0, 1, 1, 0]
far = [0, 1, 0, 0, 1]
rates = [sum(s) / len(s) * 100 for s in [near, mid, far]]
plt.bar(["3 m", "6 m", "9 m"], rates, color=["#147D78", "#E98664",
"#89B6B6"])
plt.ylabel("Success rate (%)")
plt.ylim(0, 100)
plt.show()
```

YOUR OUTPUT

Predict first. Then run your code.

The starter contains the same original 15 synthetic attempts. Changes here apply to this code experiment.

What does your chart actually say?



READ THE PICTURE

Which bar is tallest?

What does the height mean?

Every bar starts at zero.

Each group contains five shots.

In this small teaching dataset, closer shots have a higher goal rate. This does not prove that distance caused the difference.

The rate decreases as distance increases in this small synthetic dataset. This alone does not show cause.

Reference chart: original synthetic data, $n = 5$ attempts at each distance. It is separate from your edited Python experiment.

One small edit. Predict before you run.

START FROM THE SAME FIVE SHOTS

shots = [1, 0, 1, 1, 0]

Original: 3 goals ÷ 5 shots = 60%

Choose one change



Core · replace the last miss

shots[-1] = 1



Stretch · add a new goal

shots.append(1)

Predict the new rate (%)

0-100

Arithmetic simulation · the code labs run real Python.

BEFORE → AFTER

Make your prediction. Then run the arithmetic test.

Does your change replace a shot or increase the number of shots?

Your test will appear here.

The denominator changes the story.

CORE · REPLACE

`shots[-1] = 1`

One miss becomes a goal.

Does the total number of shots change?

$4 \div 5 \times 100 = 80\%$

One more goal; still five shots.

[1, 0, 1, 1, 1]

STRETCH · APPEND

`shots.append(1)`

A new successful shot is added.

What happens to both counts?

$4 \div 6 \times 100 \approx 66.7\%$

One more goal and one more shot.

[1, 0, 1, 1, 0, 1]

Append: 4 goals in 6 shots gives about 66.7%.

YOUR TURN

"My result changed because _____. The total attempts are now _____."

What if the data become more complex?

TODAY

Describe a sample.

Count goals.
Calculate rates.
See a pattern.

NEXT

Learn from examples.

Distance + angle + past outcomes
→ a model that estimates the
chance of a goal.

ALWAYS

Test the prediction.

Use new, unseen attempts.
Compare with a simple baseline.
Study the mistakes.

YOUR TURN

What else would you record besides distance?

This is a concept map, not a trained model or a performance claim.

AI can suggest. You still check.

A CLAIM TO CHECK

“Add one successful shot to 3 goals in 5 shots, and the goal rate becomes 80%.”

Teacher-written example of a claim an AI could make.

Would you trust that answer?

Check both numbers: 4 goals in 6 shots $\approx$ 66.7%.

The claim counts the new goal but forgets the new attempt. Test an answer with data before you use it.

$$4 / 6 * 100$$

You spotted what needs checking: the number of attempts changes too.

The same habit appears in real projects.

Cross-device performance of wearable sensors

QUESTION

Do devices perform equally across activities?

DATA

Readings across several physical activities

ANALYSIS

Compare results by activity

FINDING

One overall score can hide differences

Artifact: Poster Abstract · [Existing anonymized project showcase ↗](#). No numerical results are reproduced here.

YOUR TURN

How could splitting data into groups help this project, too?

Your notebook can become a poster.

DISTANCE & SCORING

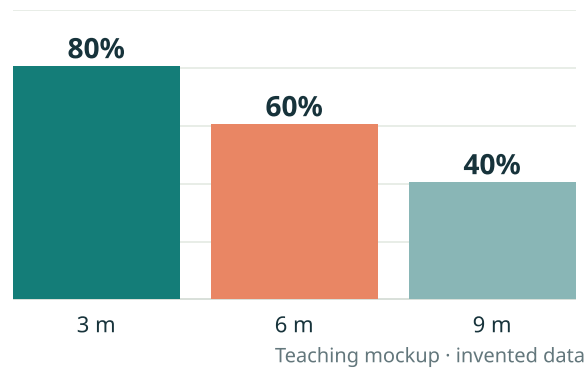
QUESTION

How does scoring vary with distance in our practice data?

LIMIT

Tiny synthetic sample.
No real-player conclusion.

Success rate · 0-100%



Build it piece by piece.

Your question

Your data and code

Your chart

Your finding and limits

YOUR TURN

Which part of this poster did you already start today?

Give a 30-second research talk.

I asked...

a question

I ran / changed...

a piece of code

I found...

a result with a number

I cannot yet claim...

a limit

YOUR TURN

Parents and partners: “What decision did you make yourself?”

Keep your evidence.

I predicted...

What did you expect to happen?

I changed...

Which one thing did you change?

I observed...

Use a number from your test.

My result cannot tell me...

What would you need more evidence for?

No challenge test yet. The download will use the original data: $3 / 5 = 60\%$.

Your note stays in this browser. No name or account needed.

What would you try next?

Can a simple rule help us make a better decision?

Next: write a rule, run more attempts, and compare what changes.

Huaxia class schedule and enrollment details will be confirmed with the school.